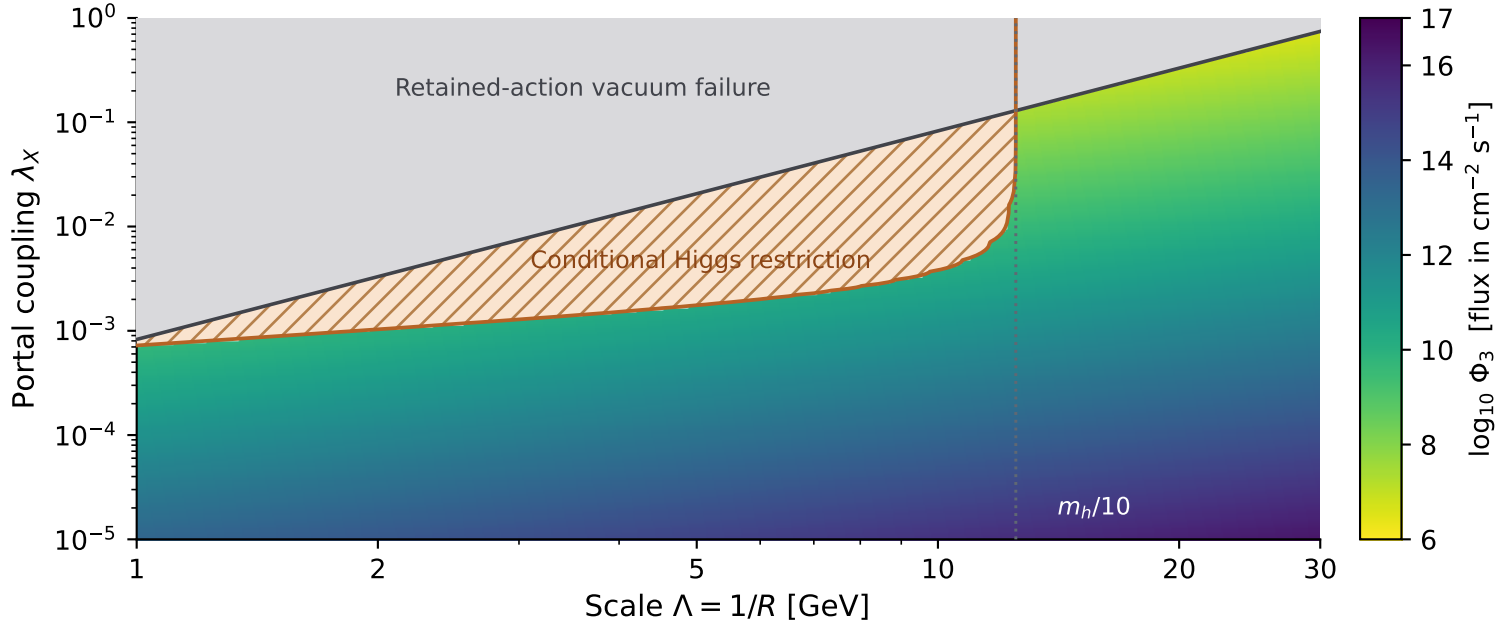


Route G2-X | Physical feasibility, not a realized detector

X-only Higgs portal / public XENON1T S2-only NR response

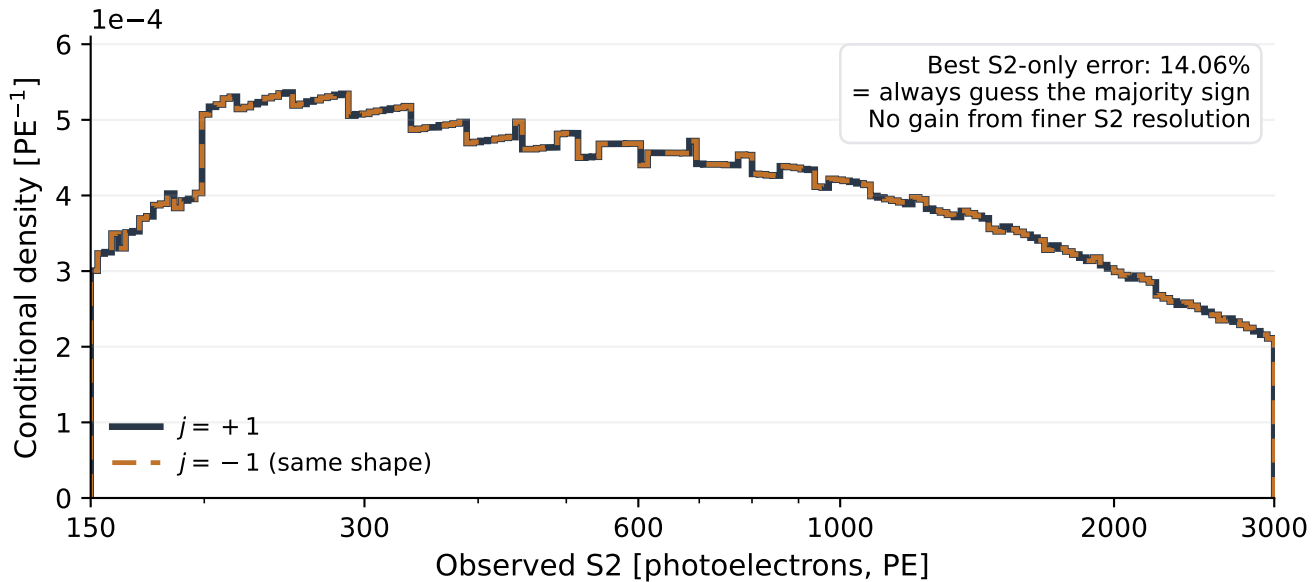
Declared scale: $m_\chi = \sqrt{26} \Lambda$; no measured radius or incident flux

A | Required flux vs conditional restrictions



Theory: $M_{D,0}^2 < 0$ ATLAS 2023: $B_{\text{inv}} < 0.107$ (95% CL)

B | Real response does not distinguish the two signs



Color = at-target flux for 3 expected accepted events, NOT a discovery threshold.

No colored region is certified viable: production, transport and backgrounds are missing.

NR window 0.7–50 keV; full S2 bins 150.027–3000 PE; exposure 356770 kg day.

Sources: XENON1T 2019 / official 2020 response; ATLAS 2023; PDG; CIAAW.

Scalar-current/Helm tree model; calibration-informed response, not new measured events.